

Investigating Hard Carbon Derived from Diverse Date Seeds for Long-Life High-Performance SIBs

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Motivation

- Global batteries **energy storage demand** expected to reach **4,700 GWh by 2030**. [1]
- Lithium-ion batteries rely on **scarce** and **expensive** materials.
- Sodium Ion Batteries (SIBs)** utilize **sodium**, which makes about **2.74% of the earth's crust**, lowering its cost significantly. [2]
- While optimal cathodes exist, the challenge lies in **identifying anodes** with sufficient Na^+ intercalation space.
- Biomass-derived hard carbon (HC)** shows great promise in many fields, especially with a **90% reliability** as **SIB anodes**.

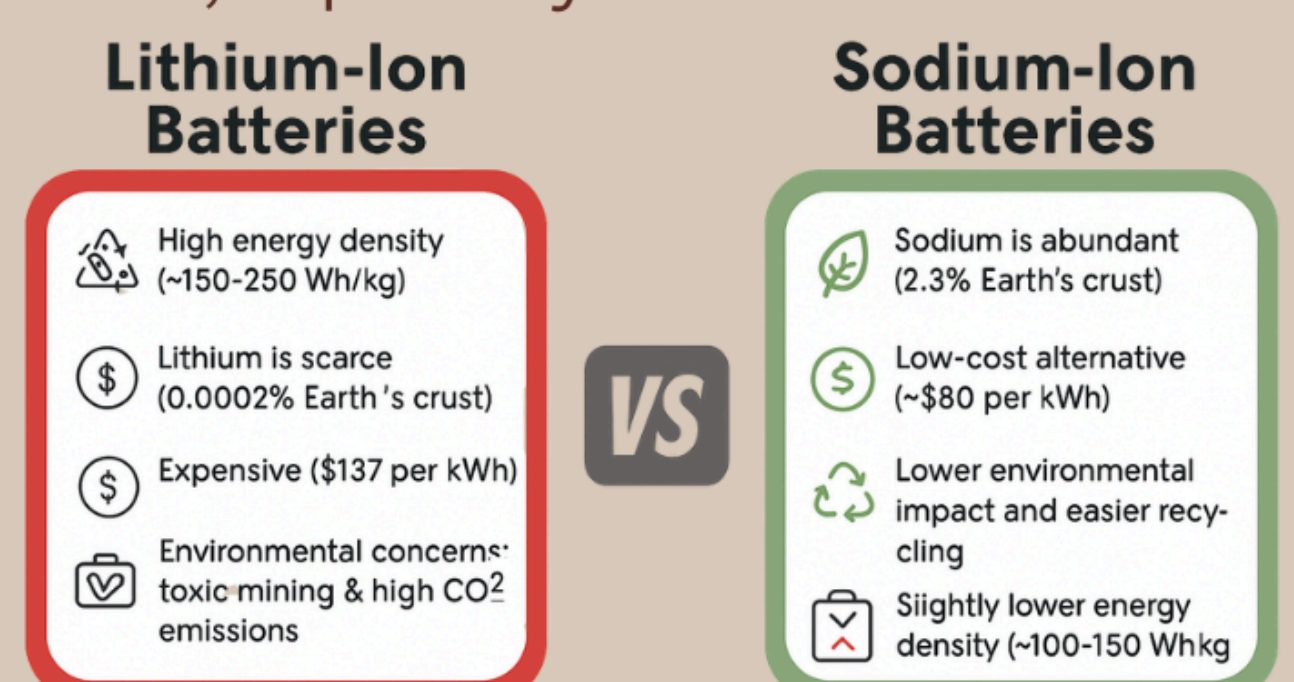


Fig. 1. Lithium-ion vs sodium-ion battery comparison.

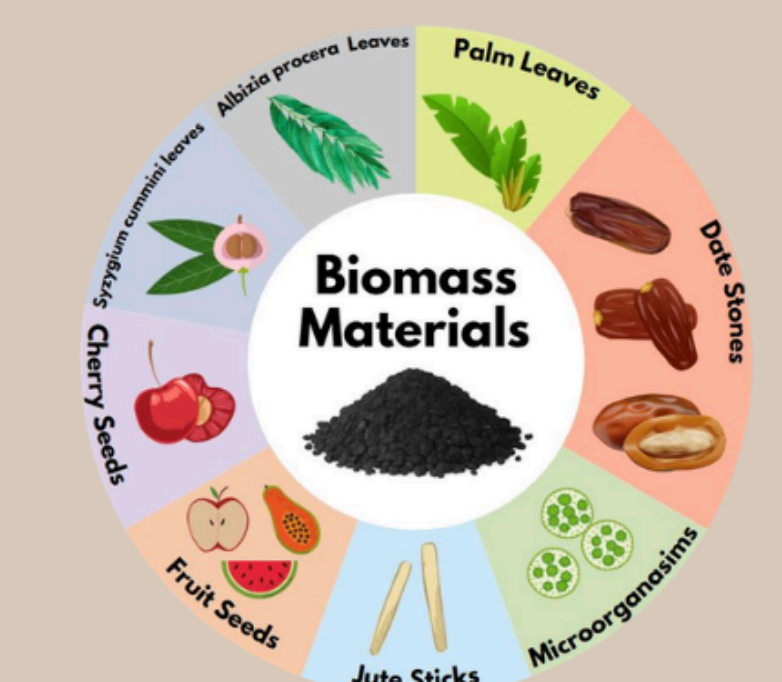


Fig. 2. Diverse biomass sources.

Background

- Hard carbons** are ideal anode materials for sodium-ion batteries due to their “**house of cards**” microstructure and expanded interlayer spacing, which **fit larger Na^+ ions**. [4]
- Sodium storage** in hard carbon involves **surface adsorption, pore filling, and intercalation** between graphene layers, enabling **stable cycling**. [5]
- Date seed waste** is an **abundant, eco-friendly** precursor for synthesizing hard carbons with tailored electrochemical properties.
- This work explores **how date seed variety** affects the **structure and performance** of hard carbon anodes.

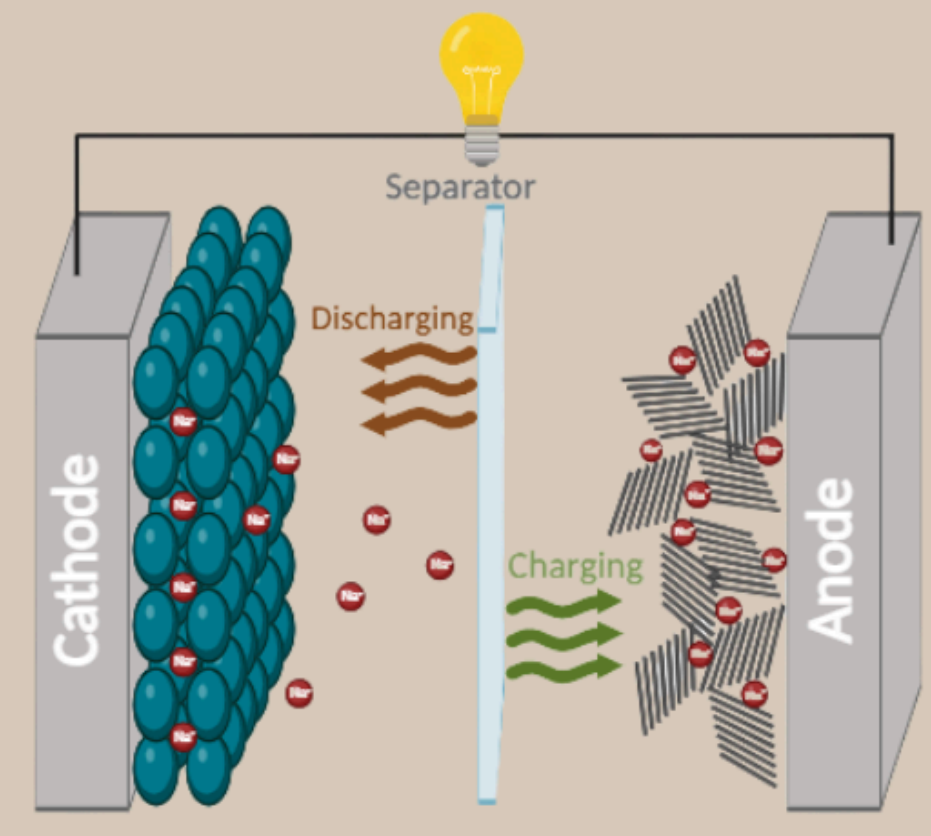


Fig. 3. SIBs with Hard Carbon Mechanism

Objectives

- Synthesize** hard carbon materials from diverse date seeds through controlled pyrolysis.
- Characterize** structural, morphological, and electrochemical properties of hard carbons
- Compare** the performance of the synthesized hard carbons for SIBs.
- Identify** the most suitable date seed precursor for high-capacity, sustainable energy storage applications.

Material Preparation

① Date Collection, Washing, and Drying



Fig. 4. Dates collection and preparation

② Powder Grinding and Filtration



Fig. 5. Fine powder filtration

Pyrolysis

③ Raw Material Annealing

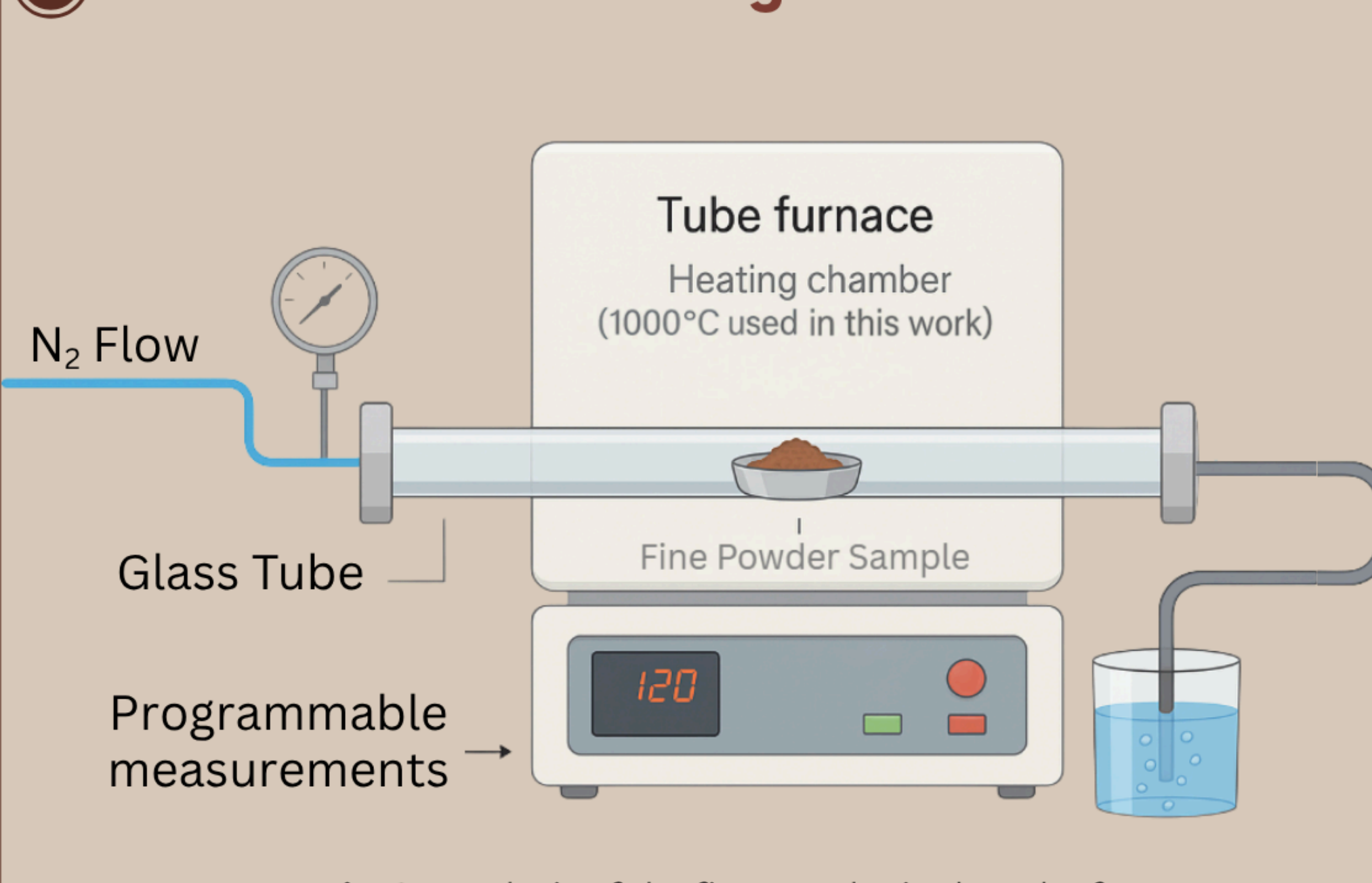


Fig. 6. Pyrolysis of the fine powder in the tube furnace

Anode Preparation

④ Washing & Drying

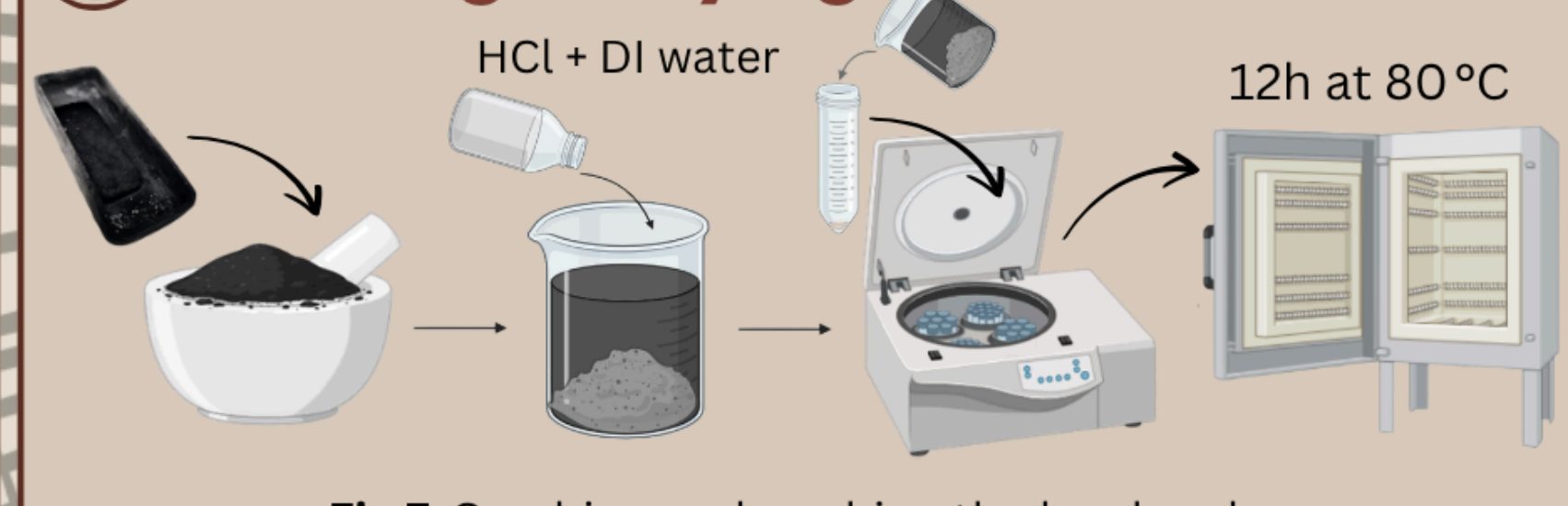


Fig. 7. Crushing and washing the hard carbon

⑤ Coating

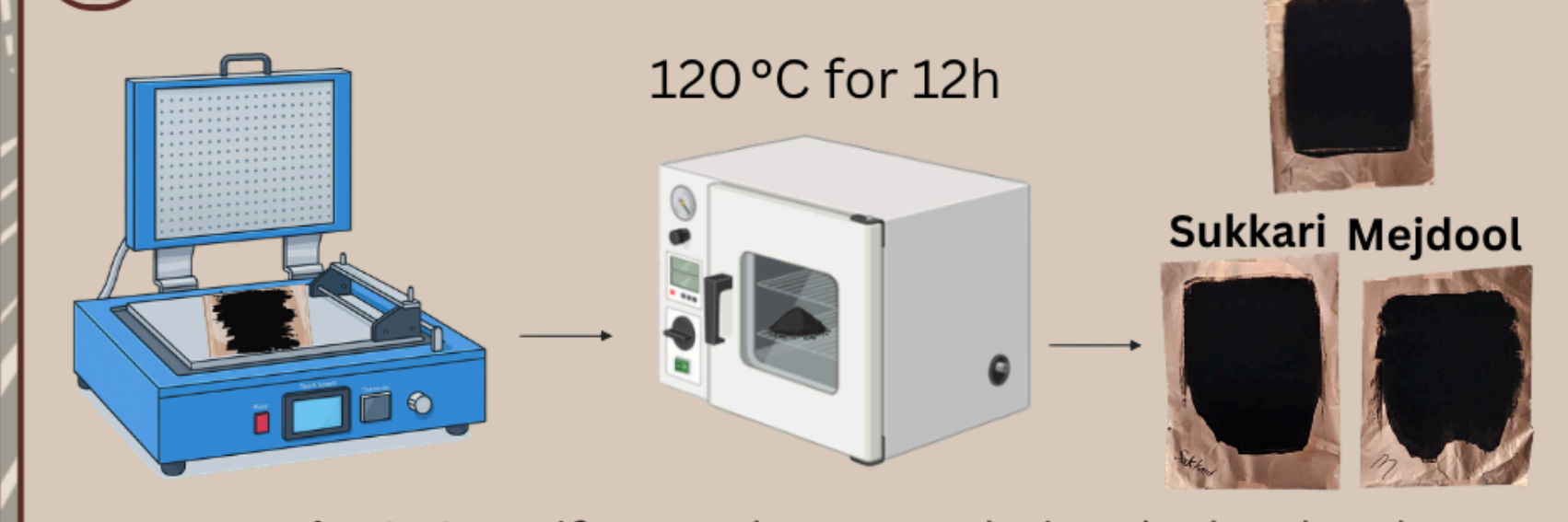


Fig. 8. Centrifuge and vacuum drying the hard carbon

Cell Synthesis

- Coin cells were prepared inside a glovebox.

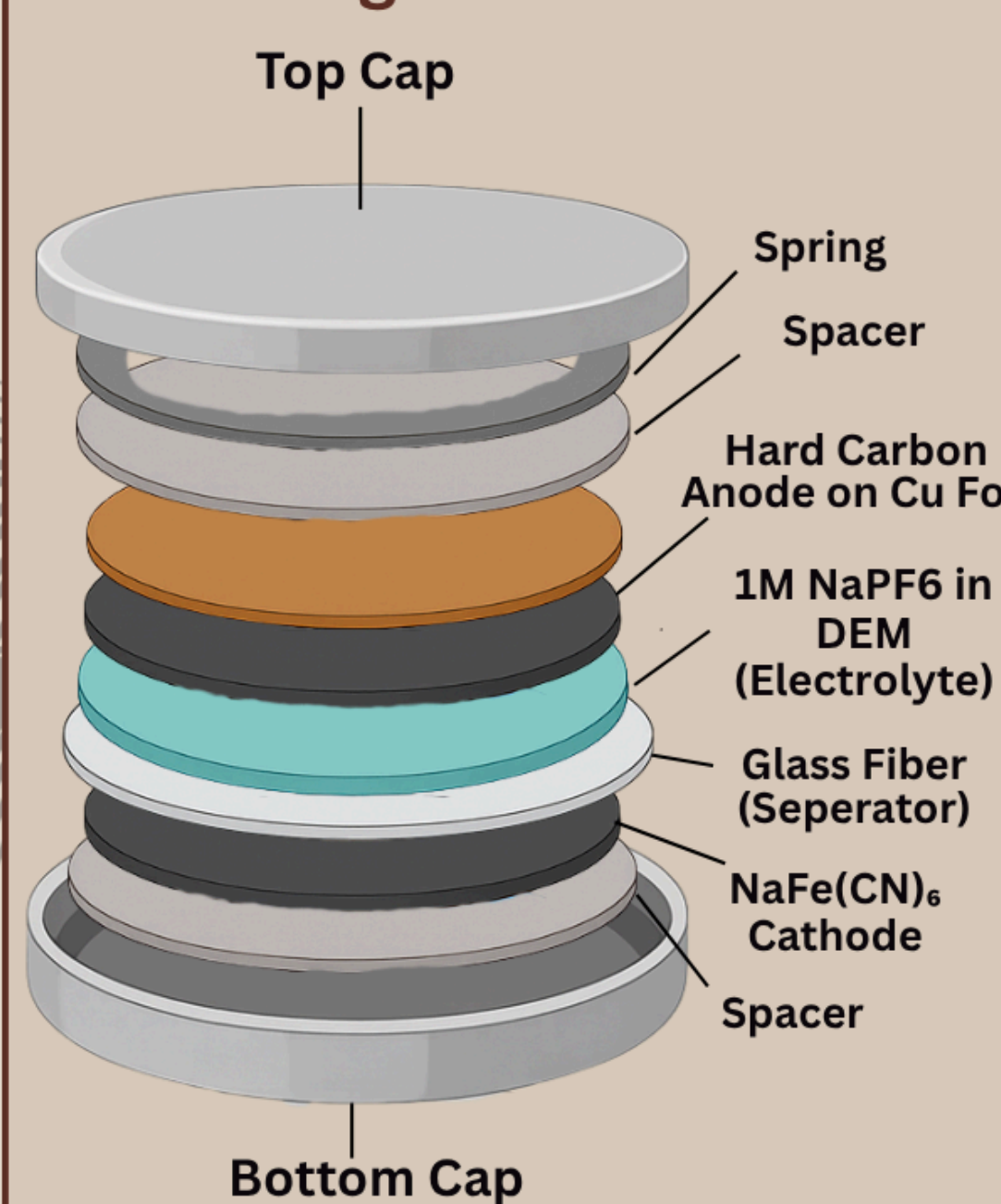


Fig. 9. Full Coin cell synthesis

Characterization (SEM, EDS, XRD, FTIR, BET)

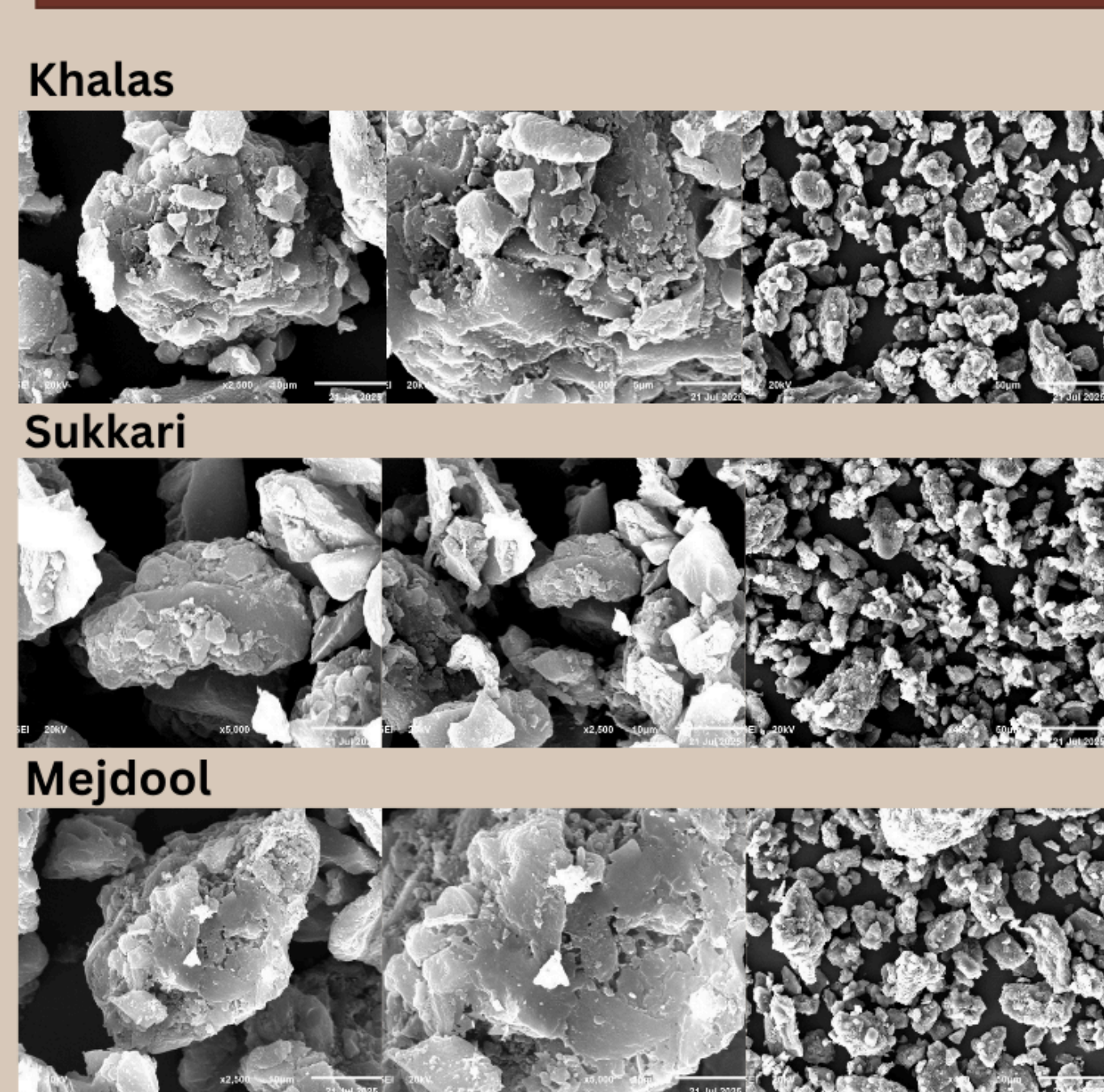


Fig. 10. SEM images of Khalas, Sukkari, and Mejdool hard carbons showing particle morphology differences.

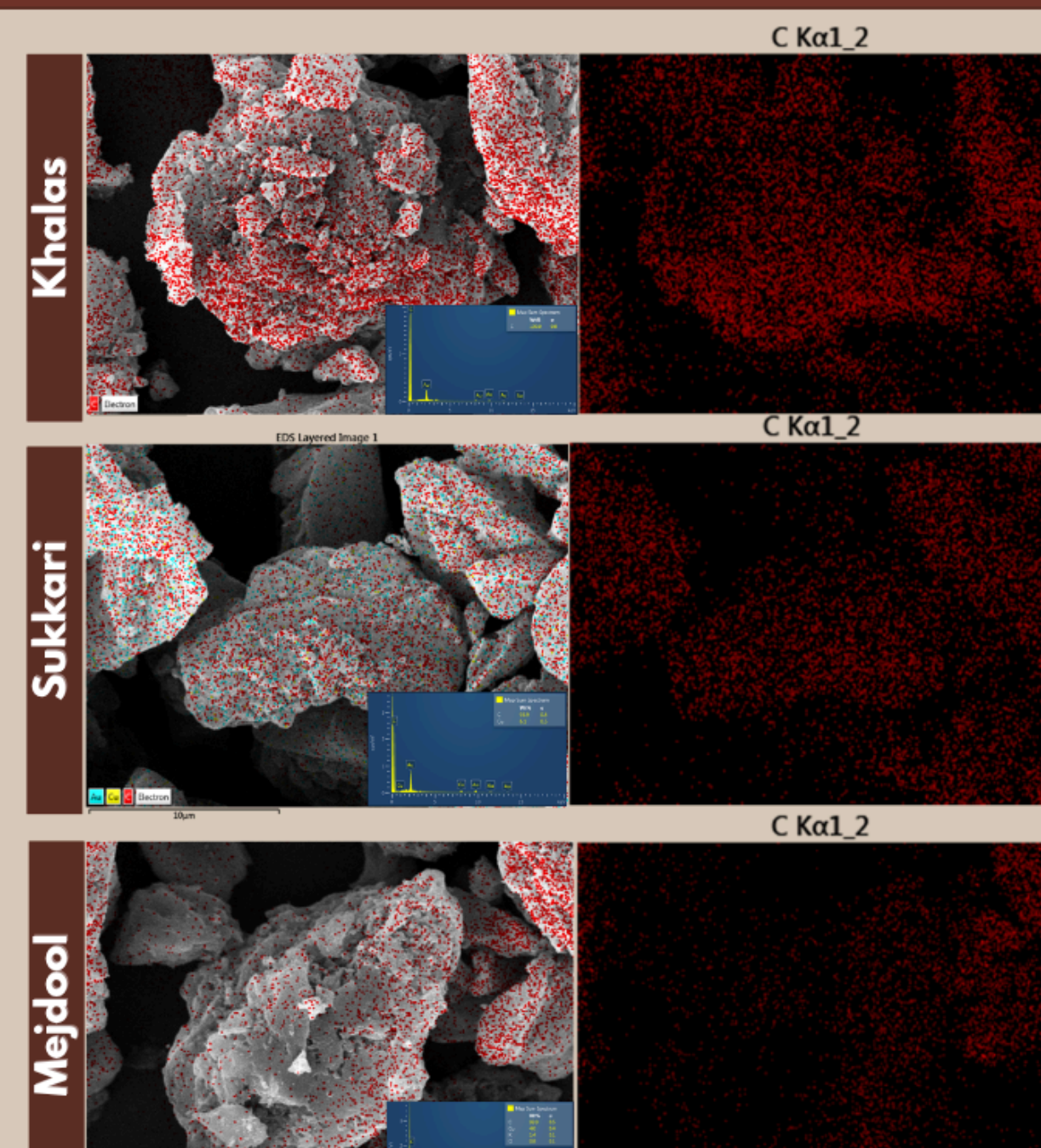


Fig. 11. EDS elemental mapping of the dates hard carbons showing uniform carbon distribution and minimal impurities across all samples.

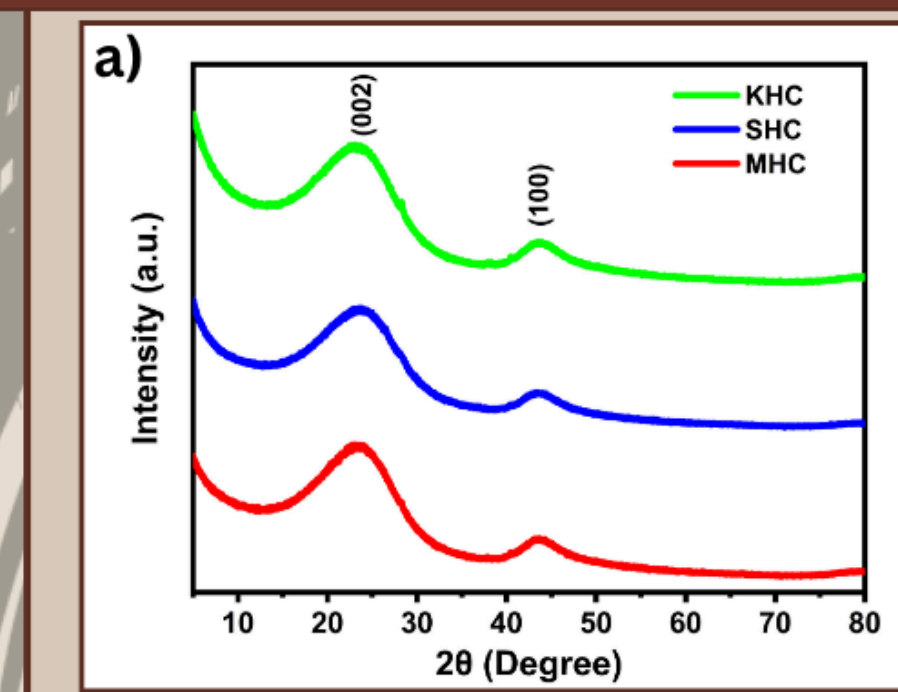


Fig. 12 (a). XRD results of the dates

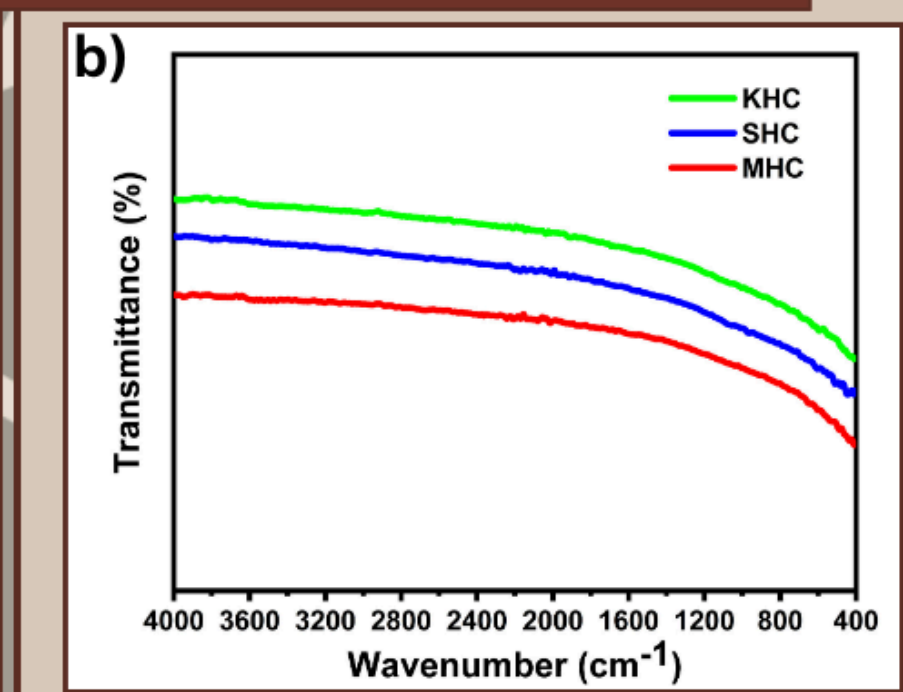


Fig. 12 (b). FTIR results of the dates

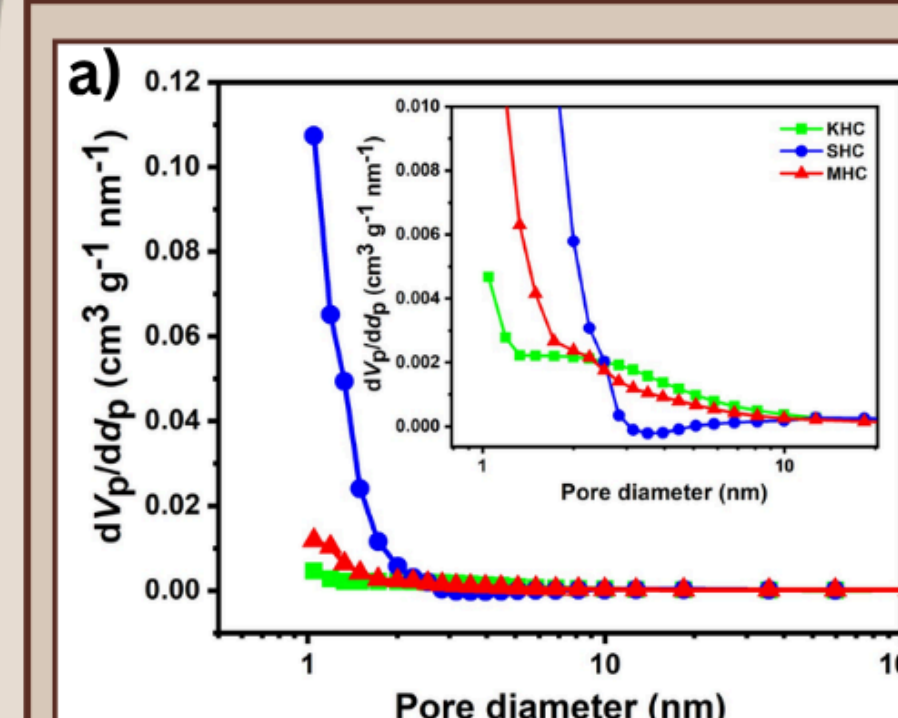


Fig. 13(a). Pore size distribution

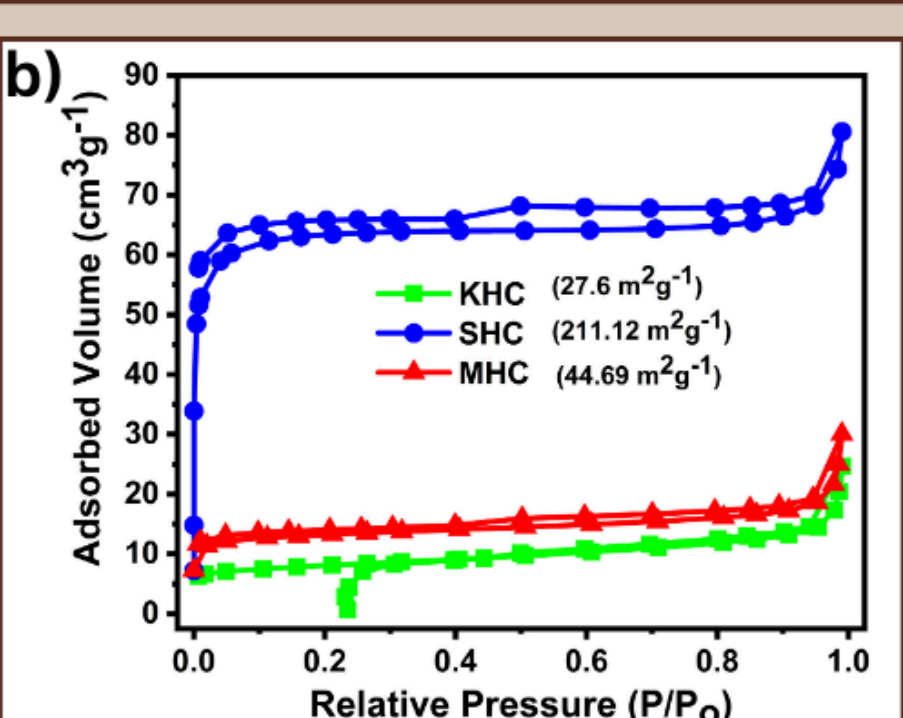


Fig. 13(b). N2 adsorption-desorption isotherms

Conclusion

Graphitic structure

Successful carbonization

Optimal Pore size

Optimal Surface Area

- All samples demonstrated **successful carbonization**, **minimal impurities** (FTIR), and **uniform carbon graphitic distribution** (EDS).
- All samples show broad peaks at (002), indicating **amorphous/disordered hard carbon**.
- Khalas, Sukkari, and Mejdool achieved surface areas of **27.6 m²/g**, **211.1 m²/g**, and **44.69 m²/g** respectively.
- Khalas, Sukkari and Mejdool demonstrated pore sizes of **1.8 nm**, **4.8 nm**, and **6.3 nm** respectively.
- These dates illustrate **great potential** for SIBs applications.

Future Work

Long Life Stability

Capacity determination at various current densities

1

2

3

4

Elemental analysis of solid electrolyte interface

Economical analysis

Applications



Grid Storage



Renewables



Electric Vehicles



Electronics

References

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